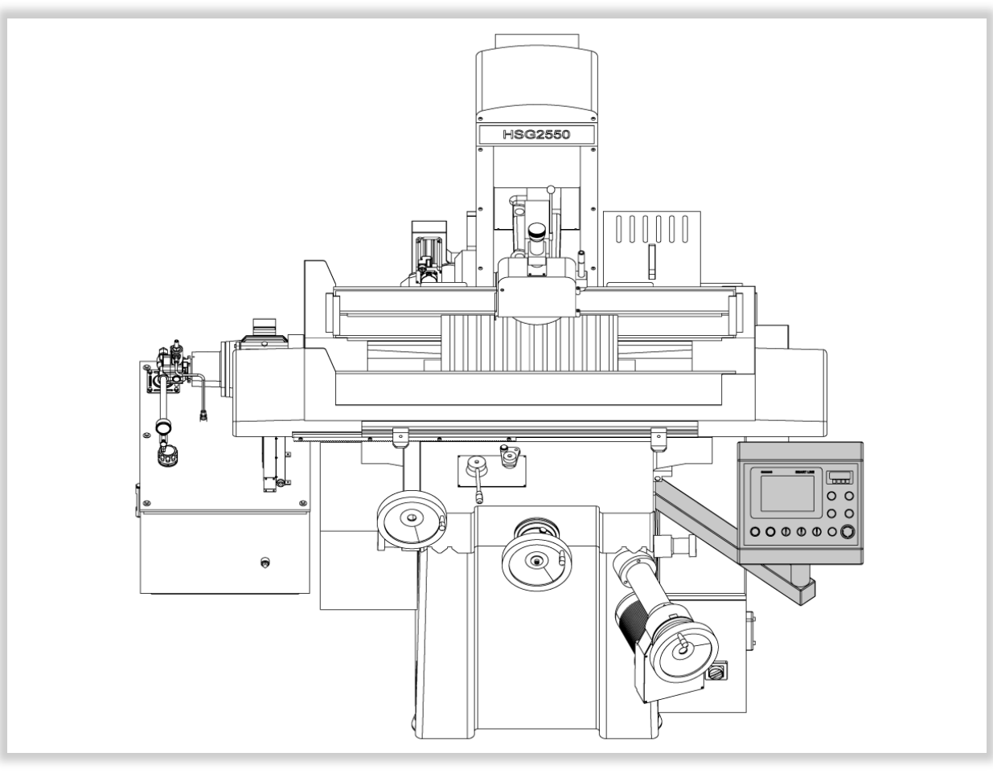
5.1 HSG - OPERATION

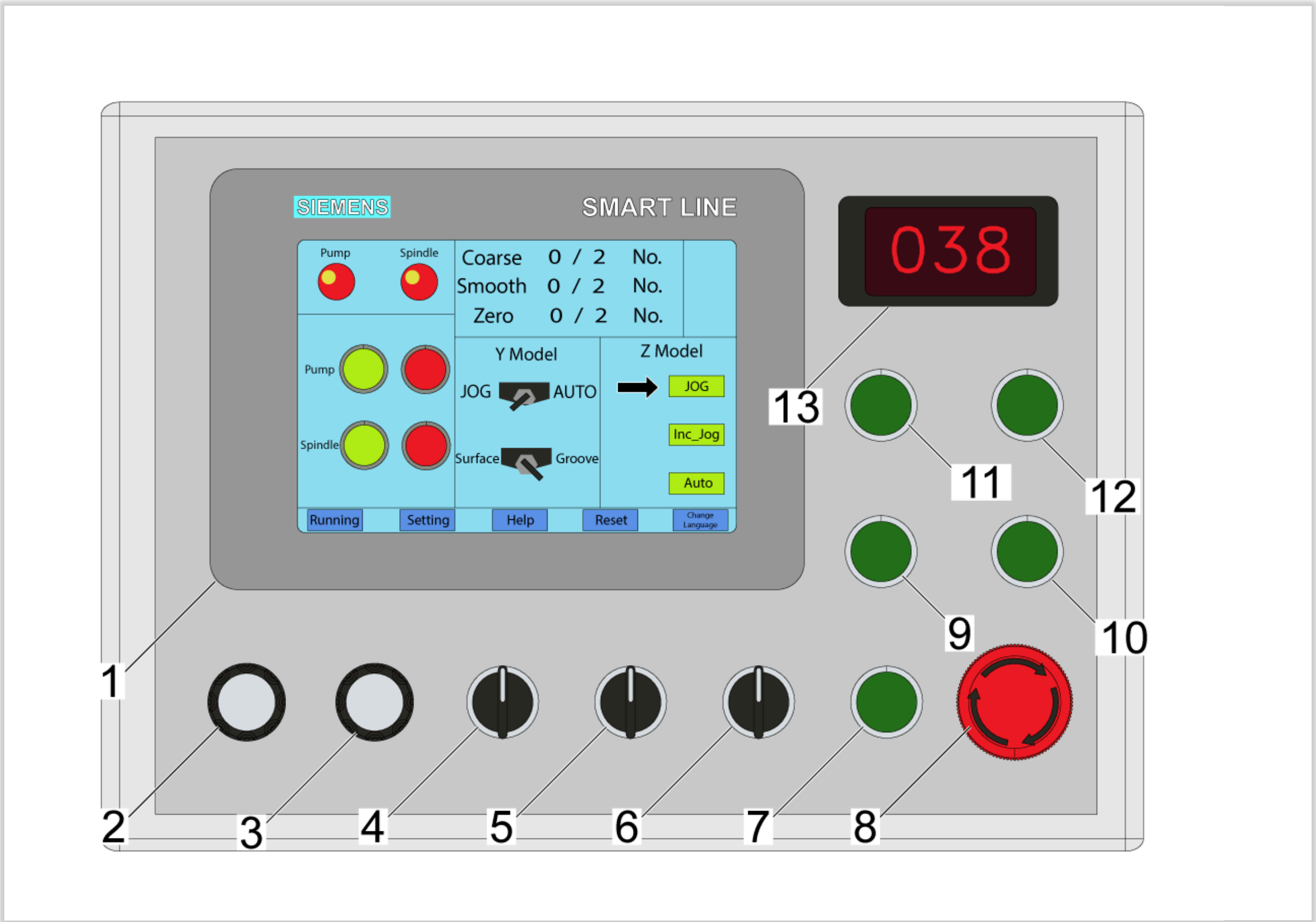
Haas Surface Grinder (HSG) - Operation

Overview



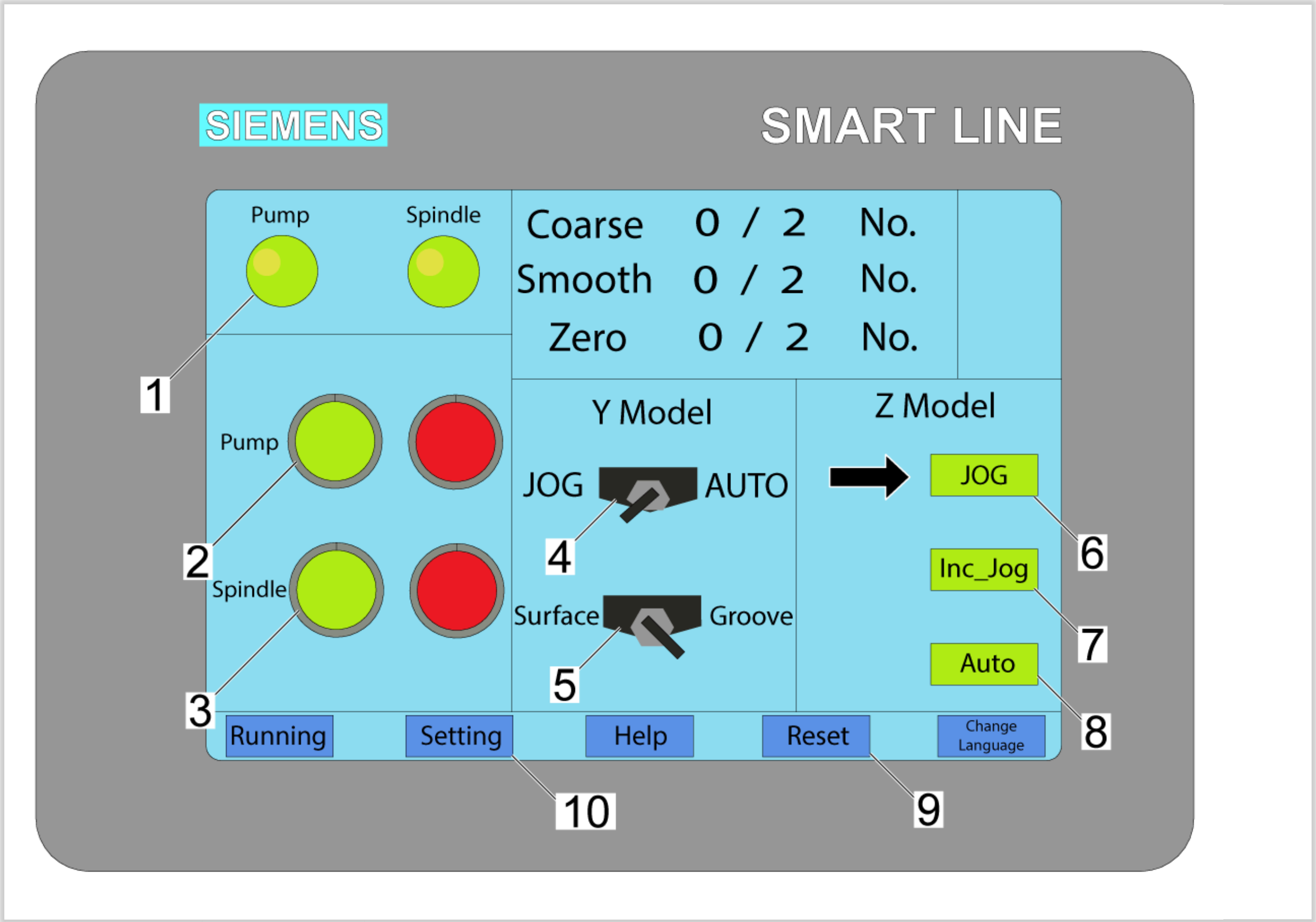
This chapter will provide details on how to operate the surface grinder. It will cover how to navigate the main control as well as other basic operation details.

Main Control


SIEMENS
SMART LINE
Pump
Spindle
Coarse 0 / 2 No.
Smooth 0 / 2 No.
Zero 0 / 2 No.
Y Model
Z Model
JOG
AUTO
Surface
Groove
Inc_Jog
Auto
Running
Setting
Help
Reset
Change Language
038
1
2
3
4
5
6
7
8
9
10
11
12
13

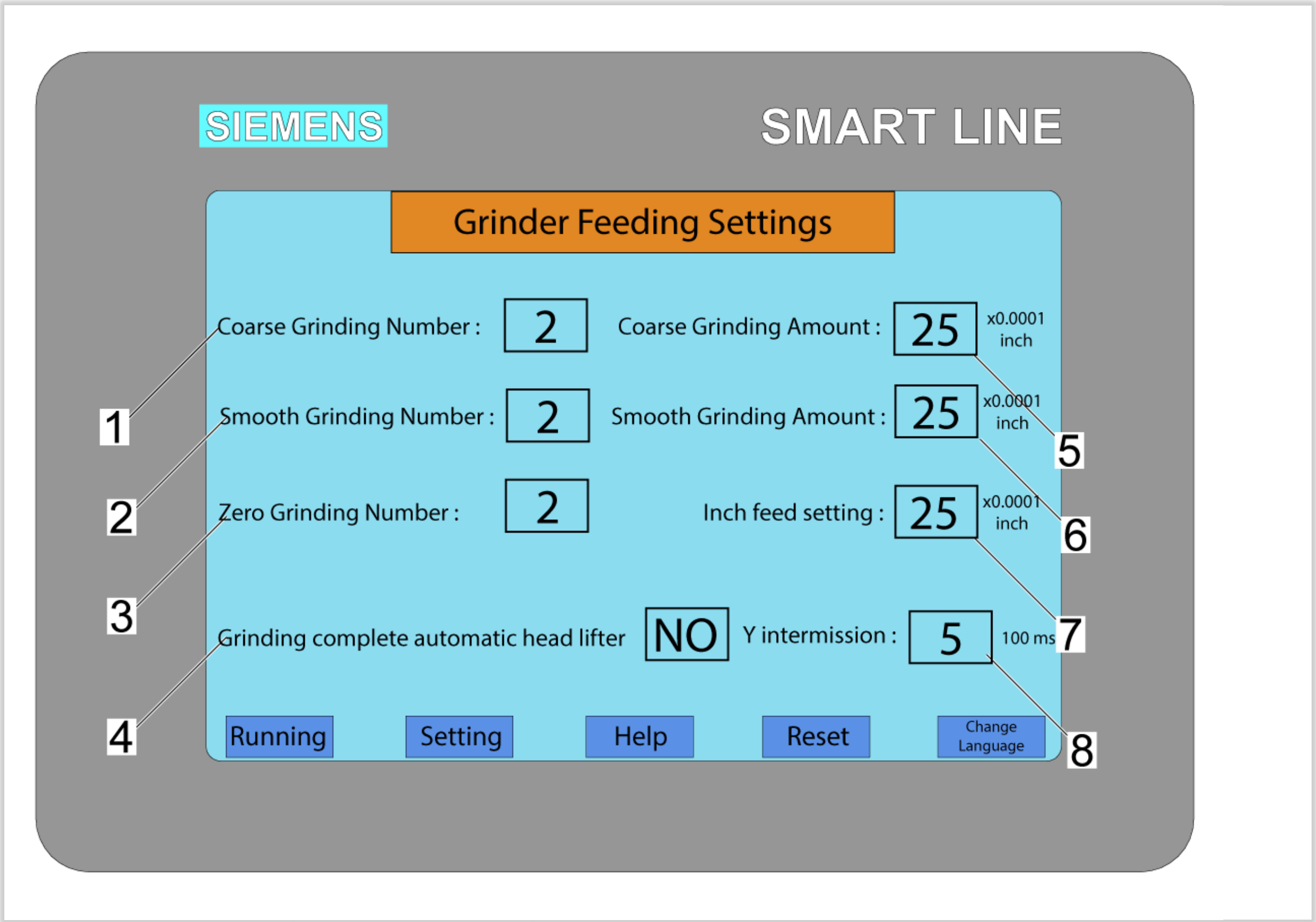
NUMBER	TYPE	NAME
1	Screen	Main screen
2	Dials	Electro-magnetic chuck voltage
3		Y-axis intermittent feed
4	Switches	Work lamp on/off
5		Coolant pump on/off
6		Electro-magnetic chuck on/off
7	Buttons	Power up
8		E-Stop
9		Z-axis rapid up
10		Z-axis rapid down
11		Y-axis rapid forward
12		Y-axis rapid backwards
13	Display	Magnetic chuck display

Running Page


SIEMENS
SMART LINE
Pump
Spindle
Coarse 0 / 2 No.
Smooth 0 / 2 No.
Zero 0 / 2 No.
Pump
Spindle
Y Model
JOG
AUTO
Surface
Groove
Z Model
JOG
Inc_Jog
Auto
Running
Setting
Help
Reset
Change Language
1
2
3
4
5
6
7
8
9
10

NUMBER	TYPE	NAME
1	Screen	Pump/Spindle On/Off indicator lights
2		Pump On/Off buttons
3		Spindle On/Off buttons
4		Y-Axis mode selection (Jog or Auto)
5		Y-Axis mode selction (Surface / Groove)
6		Z-Axis Jog mode buton
7		Z-Axis Incremental Jog mode button (Controlled usig inch feed setting)
8		Z-Axis Auto mode button (Controlled using settings page)
9		Reset button (used after automatic mode has completed its cycle)
10		Setting button

Settings Page


SIEMENS
SMART LINE
Grinder Feeding Settings
Running
Setting
Help
Reset
Change Language
1
2
3
4
5
6
7
8

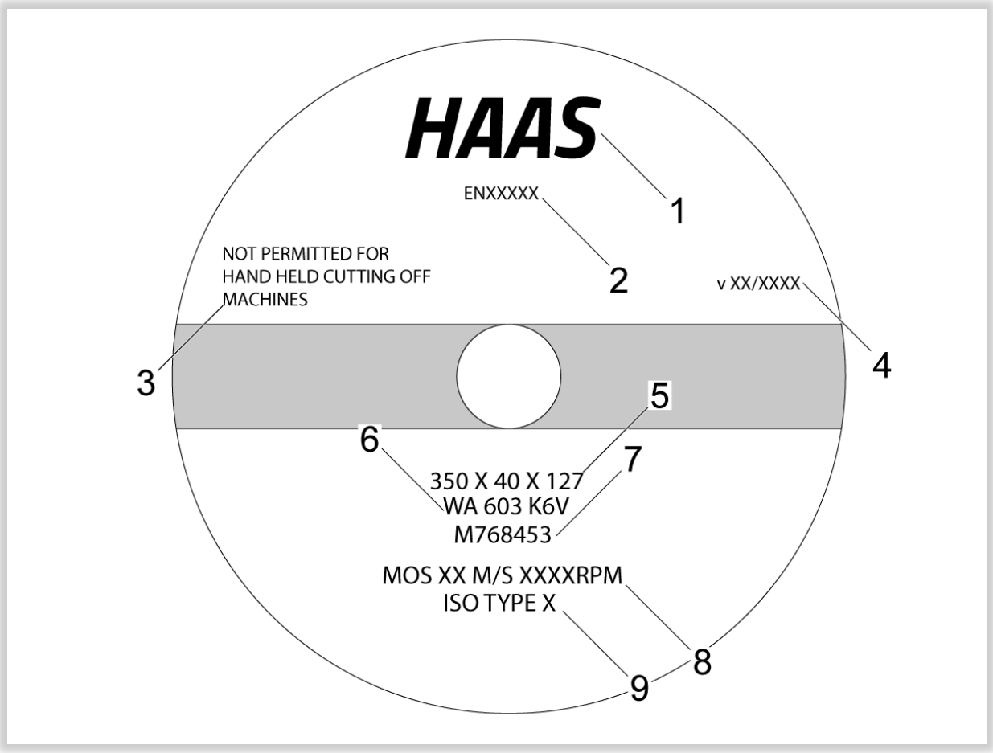
Coarse Grinding Number : 2
Coarse Grinding Amount : 25 x0.0001 inch
Smooth Grinding Number : 2
Smooth Grinding Amount : 25 x0.0001 inch
Zero Grinding Number : 2
Inch feed setting : 25 x0.0001 inch
Grinding complete automatic head lifter NO
Y intermission : 5 100 ms

NUMBER	TYPE	NAME
1	Screen	Number of Coarse grinding passes
2		Number of Smooth grinding passes
3		Number of zero (no grinding) passes
4		Automatic head lifter On/Off
5		Coarse grinding cutting amount
6		Smooth grinding cutting amount
7		Z-axis incremental job amount
8		Y-axis movement time

5.2 HSG - GRINDING WHEEL IDENTIFICATION

Haas Surface Grinder (HSG) - Grinding Wheel Identification

Grinding wheel Identification


HAAS
ENXXXXX
NOT PERMITTED FOR HAND HELD CUTTING OFF MACHINES
v XX/XXXX
350 X 40 X 127
WA 603 K6V
M768453
MOS XX M/S XXXXRPM
ISO TYPE X
1
2
3
4
5
6
7
8
9

1. Trade Mark
2. Test Record
3. Restriction of Use
4. Expiration Date
5. Dimensions in mm
6. Specification Mark
7. Code Number
8. Maximum Operating Speed
9. Iso Type No (Shape)

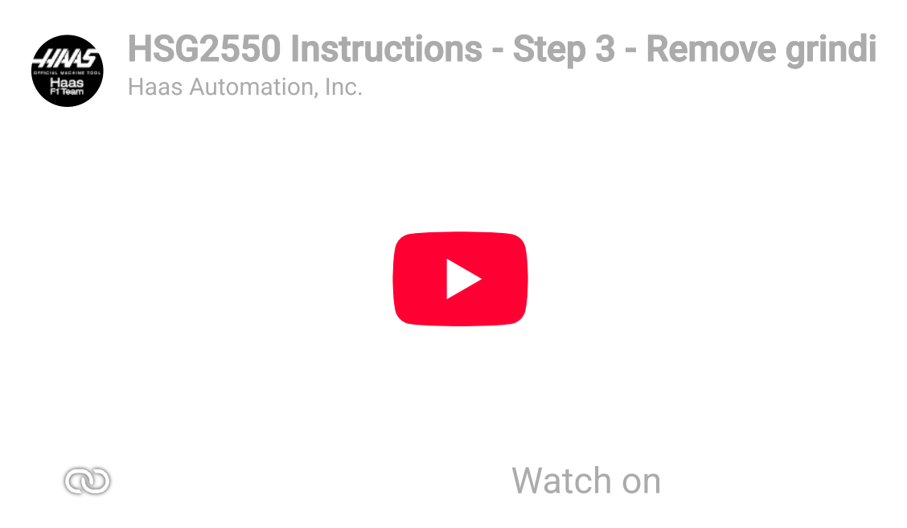
Grinding Wheel Abrasives

ABRASIVE	ABRASIVE DENOTATION
Brown Aluminum Oxide	A
White Aluminum Oxide	WA
Mixture of brown and white Aluminum Oxide	DA or MA
Pink Aluminum Oxide (dark)	PA
Pink Aluminum Oxide (light)	PWA
Black Silicone Carbide	C
Green Silicone Carbide	GC
Combination of black and green Silicone Carbide	CGC
Blend of Aluminum Oxide and Silicone Carbide	AC
Zirconia with Brown Aluminum Oxide	ZA
Micro Crystalline Grains	MCA
Ruby Grains	RB
Semi friable Aluminum Oxide	FS
Sol Gel Grain	CG

5.3 HSG - GRINDING WHEEL REPLACEMENT AND BALANCING

Haas Surface Grinder (HSG) - Grinding Wheel Replacement and Balancing

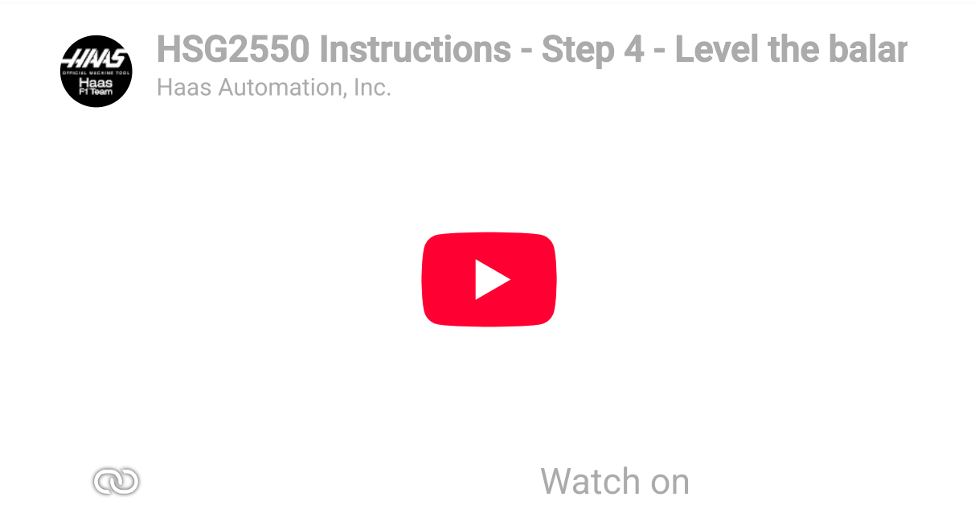
Replacement


HSG2550 Instructions - Step 3 - Remove grindi
Haas Automation, Inc.

1
Remove the grinding wheel from the machine using the provided wheel removal tool.

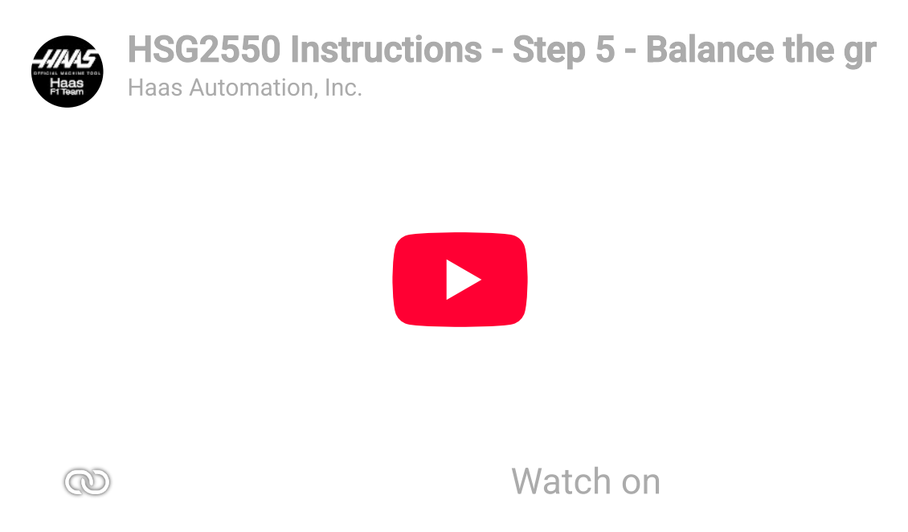
Screw on the removal bolt in the clockwise direction until the grinding wheel is off the taper.

Note : The nut thread is left handed so ensure that the wrench is rotated clockwise.


HSG2550 Instructions - Step 4 - Level the balar
Haas Automation, Inc.

2
Clean all mating surfaces and place the the wheel balancing base and height gauge on top of a level granite table.

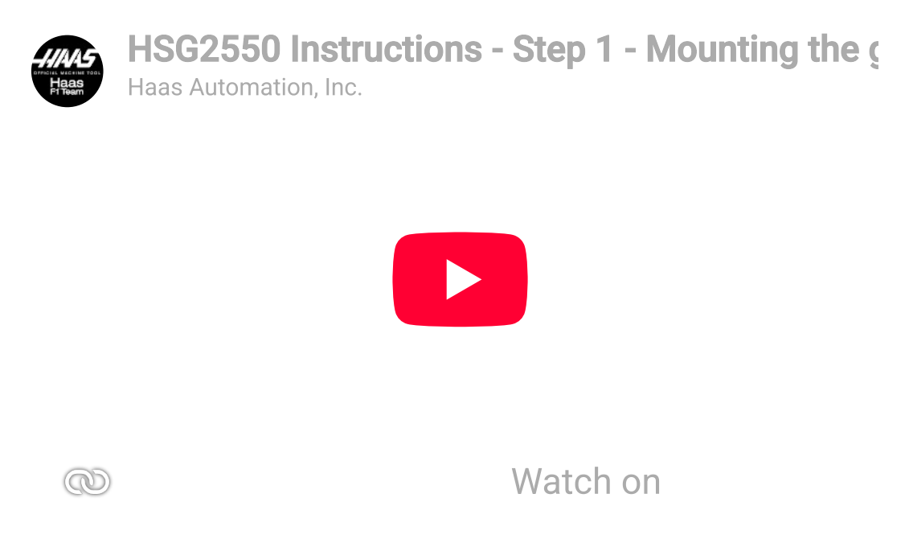
Sweep an indicator across the high point of the balancing rail and adjust the bottom 2 knobs untl all 4 corners read withing 0.0001"(0.0025mm).


HSG2550 Instructions - Step 5 - Balance the gr
Haas Automation, Inc.

3
Place the tapered arbor firmly into the grind wheel hub. Place the arbor onto the balancing rails and monitor the rotation of the wheel then let it come to rest.

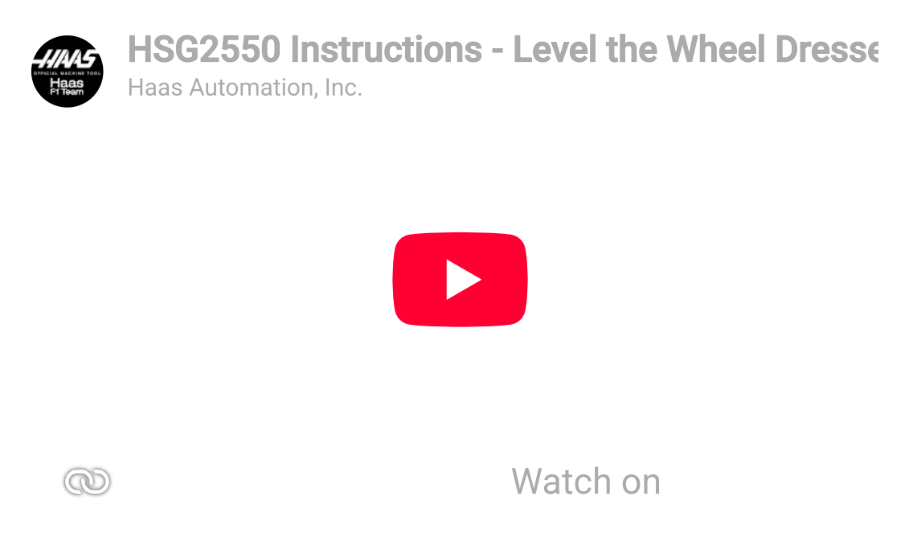
Mark the heavy and light points on the wheel using chalk and add a balance weight to the light side of the wheel. Equally space and place the other 2 balance weight around the wheel hub.

Lift and rotate the wheel 90 degrees and rest onto the balancing rails to observe movement. Adjust the location of the weight accordingly until the wheel is able to be placed on all 4 quadrants without movement,


HSG2550 Instructions - Step 1 - Mounting the g
Haas Automation, Inc.

4
Place wheel hub through rear bore of the grind wheel then place the hub disk on the opposing side and tighten the hub nut.

Clean the spindle taper then place the wheel firmly on it and tighten the wheel nut in the anti-clockwise direction.


HSG2550 Instructions - Level the Wheel Dresse
Haas Automation, Inc.

5
Use the handle to manually move the diamond pen on top of the grind wheel.

Rotate the adjustment knob until the pen touches the grinding wheel then return the ram off of the wheel.

Adjust the knob then pull the handle and feed the diamond per fully accross the grind wheel.

Repeat until the grind wheel is fully dressed.